Day 05 – Piscine Java

SQL/JDBC

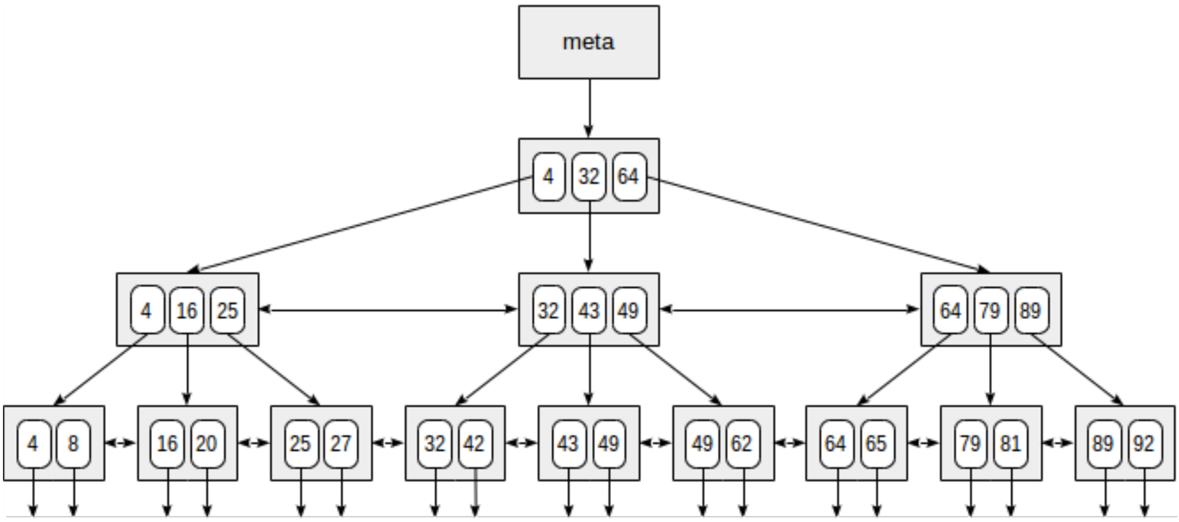
> *Today you will use the key mechanisms to work with PostgreSQL DBMS via JDBC.*

Contents

Preamble

As you know, relational databases consist of a set of linked tables. Each table has a set of rows and columns. If there is a huge number of rows in a table, searching data with a specific value in the column may obviously take a lot of time.

To solve this problem, up-to-date DBMS use an index mechanism. **BTree** data structure is an implementation of index concept.



This index may be used for a column of a table. Since a tree is always balanced, searching for any value takes the same amount of time.

Rules that substantially expedite a search are as follows:

1. Keys in each node are ordered.
2. Root contains **1** to **t-1** keys.
3. Any other node contains **t-1** to **2t-1** keys.
4. If a node contains **k1, k2, … kn** keys, it has **n+1** derived classes.
5. The first derived class and all its derived classes contain keys that are less than or equal to **k1**.
6. The last derived class and all its derived classes contain keys that are greater than or equal to **kn**.
7. For **2 <= i <= n**, **i**-th derived class and all its derived classes contain keys in the **(ki-1, ki)** range.

Therefore, to search for a value, you just need to determine which derived class to go down to. This allows to avoid looking through the entire table.

This approach apparently has a lot of specifics. For example, if new values constantly go into a table, DBMS will always rebuild the index which will slow down the system.

General Rules

- Use this page as the only reference. Do not listen to any rumors and speculations about how to prepare your solution.
- Now there is only one Java version for you, **1.8**. Make sure that compiler and interpreter of this version are installed on your machine.

- You can use IDE to write and debug the source code.
- The code is read more often than written. Read carefully the document where code formatting rules are given. When performing each task, make sure you follow the generally accepted Oracle standards.
- Comments are not allowed in the source code of your solution. They make it difficult to read the code.
- Pay attention to the permissions of your files and directories.
- To be assessed, your solution must be in your GIT repository.
- Your solutions will be evaluated by your piscine mates.
- You should not leave in your `src` directory any other file than those explicitly specified by the exercise instructions. It is recommended that you modify your `.gitignore` to avoid accidents.
- When you need to get precise output in your programs, it is forbidden to display a precalculated output instead of performing the exercise correctly.
- Have a question? Ask your neighbor on the right. Otherwise, try with your neighbor on the left.
- Your reference manual: mates / Internet / Google. And one more thing. There's an answer to any question you may have on Stackoverflow. Learn how to ask questions correctly.
- Read the examples carefully. They may require things that are not otherwise specified in the subject.
- Use `System.out` for output.
- And may the Force be with you!
- Never leave that till tomorrow which you can do today ;)

Rules of the Day

- Use **PostgreSQL** DBMS in all tasks.
- Connect the up-to-date version of **JDBC driver**.
- To interact with the database, you may use classes and interfaces of the `java.sql` package (implementations of corresponding interfaces will be automatically included from the archive containing the driver).

Exercise 00 – Tables & Entities

Exercise 00: Tables & Entities	
Turn-in directory	`ex00`
Files to turn-in	`Chat-folder`

Throughout this week, we will be implementing Chat functionality. In this chat, user can create or choose an existing chatroom. Each chatroom can have several users exchanging messages.

Key domain models which both SQL tables and Java classes must be implemented for are:

User:
- User ID
- Login
- Password
- List of created rooms
- List of chatrooms where a user socializes

Chatroom:
- Chatroom ID
- Chatroom name
- Chatroom owner
- List of messages in a chatroom

Message:
- Message ID
- Message author
- Message room
- Message text
- Message date/time

Create `schema.sql` file where you will describe `CREATE TABLE` operations to create tables for the project. You should also create `data.sql` file with test data `INSERT`s (at least five in each table).

Important requirement:

Let's assume that `Course` entity has a one-to-many relationship with `Lesson` entity. Their object-oriented relation should then look as follows:

```
class Course {
    private Long id;
    private List<Lesson>
lessons; // there are numerous lessons in the course
     ...
}

class Lesson {
    private Long id;
    private Course course; // the lesson contains a course it is
linked to
     ...
}
```

Additional requirements:

- To implement relational links, use **one-to-many** and **many-to-many** link types.
- Identifiers should be numeric.
- Identifiers shall be generated by DBMS.
- `equals()`, `hashCode()` and `toString()` shall be redefined correctly inside Java classes.

Exercise project structure:

- **Chat**
    - `src`
        - `main`
            - `java`
                - `edu.school21.chat`
                    - `models` — domain knowledge models
            - `resources`
                - `schema.sql`
                - `data.sql`
    - `pom.xml`

Exercise 01 – Read/Find

Exercise 01: Read/Find	
Turn-in directory	`ex01`
Files to turn-in	`Chat-folder`

Data Access Object (DAO, Repository) is a popular design template that allows to separate key business logic from data handling logic in an application.

Let's assume that we have an interface called `CoursesRepository` which provides access to course lessons. This interface may look as follows:

```
public interface CoursesRepository {
    Optional<Course> findById(Long id);
    void delete(Course course);
    void save(Course course);
    void update(Course course);
    List<Course> findAll();
}
```

You need to implement `MessagesRepository` with a **SINGLE** `Optional<Message> findById(Long id)` method and its `MessagesRepositoryJdbcImpl` implementation.

This method shall return a `Message` object where author and chatroom will be specified. In turn, there is **NO NEED** to enter subentities (list of chatrooms, chatroom creator, etc.) for the author and the chatroom.

The implemented code must be tested in `Program.java` class. Example of the program operation is as follows (the output may differ):

```
$ java Program
Enter a message ID
-> 5
Message : {
  id=5,

author={id=7,login="user",password="user",createdRooms=null,rooms=null
},
  room={id=8,name="room",creator=null,messages=null},
  text="message",
  dateTime=01/01/01 15:69
}
```

Exercise project structure:

- **Chat**
  - `src`
    - `main`
      - `java`
        - `edu.school21.chat`
          - `models` — domain knowledge models
          - `repositories` — repositories
          - `app`
            - `Program.java`
      - `resources`
        - `schema.sql`
        - `data.sql`
  - `pom.xml`

`MessagesRepositoryJdbcImpl` shall accept `DataSource` interface of `java.sql` package as a constructor parameter. For `DataSource` implementation, use **HikariCP** library — a pool of connections to the database which considerably expedites the use of storage.

Exercise 02 – Create/Save

Exercise 02: Create/Save	
Turn-in directory	`ex02`
Files to turn-in	`Chat-folder`

Now you need to implement `save(Message message)` method for `MessagesRepository`.

Thus, we need to define the following subentities for the entity we are saving — a message author and a chatroom. It is also important to assign IDs that exist in the database to chatroom and author.

Example of save method use:

```
public static void main(String args[]) {
     ...
    User creator = new User(7L, "user", "user", new ArrayList(), new
ArrayList());
    User author = creator;
    Room room = new Room(8L, "room", creator, new ArrayList());
    Message message = new Message(null, author, room, "Hello!",
LocalDateTime.now());
    MessagesRepository messagesRepository = new
MessagesRepositoryJdbcImpl( ... );
    messagesRepository.save(message);
    System.out.println(message.getId()); // ex. id == 11
}
```

So, `save` method shall assign ID value for the incoming model after saving data in the database. If author and room have no ID existing in the database assigned, or these IDs are null, throw `RuntimeException` `NotSavedSubEntityException` (implement this exception on your own).

Test the implemented code in `Program.java` class.

Exercise 03 – Update

Exercise 03: Update	
Turn-in directory	`ex03`
Files to turn-in	`Chat-folder`

Now we need to implement `update` method in `MessageRepository`. This method shall fully update an existing entity in the database. If a new value of a field in an entity being updated is `null`, this value shall be saved in the database.

Example of update method use:

```
public static void main(String args[]) {
    MessagesRepository messagesRepository = new
MessagesRepositoryJdbcImpl( ... );
    Optional<Message> messageOptional =
messagesRepository.findById(11);
    if (messageOptional.isPresent()) {
        Message message = messageOptional.get();
        message.setText("Bye");
        message.setDateTime(null);
        messagesRepository.update(message);
    }
     ...
}
```

In this example, the value of the column storing the message text will be altered, whereas message time will be `null`.

Exercise 04 – Find All

Exercise 04: Find All	
Turn-in directory	`ex04`
Files to turn-in	`Chat-folder`

Now you need to implement `UsersRepository` interface and `UsersRepositoryJdbcImpl` class using a **SINGLE** `List<User> findAll(int page, int size)` method.

This method shall return `size` users shown in the page with page number `page`. This "piecewise" data retrieval is called **pagination**. Thus, DBMS divides the overall set into pages each containing `size` entries. For example, if a set contains 20 entries with `page = 3` and `size = 4`, you retrieve users 12 to 15 (user and page numbering starts from 0).

The most complicated situation in converting relational links into object-oriented links happens when you retrieve a set of entities along with their subentities. In this task, each user in the resulting list shall have included dependencies — a list of chatrooms created by that user, as well as a list of chatrooms the user participates in.

Each subentity of the user **MUST NOT** include its dependencies, i.e. list of messages inside each room must be empty.

The implemented method operation should be demonstrated in `Program.java`.

> **Notes:**
> - `findAll(int page, int size)` method shall be implemented by a **SINGLE** database query. It is not allowed to use additional SQL queries to retrieve information for each user.
> - We recommend using **CTE PostgreSQL**.
> - `UsersRepositoryJdbcImpl` shall accept `DataSource` interface of `java.sql` package as a constructor parameter.